Wind Runner

MIE 540 Group 8:

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Market Gap & Opportunity

Market Structure

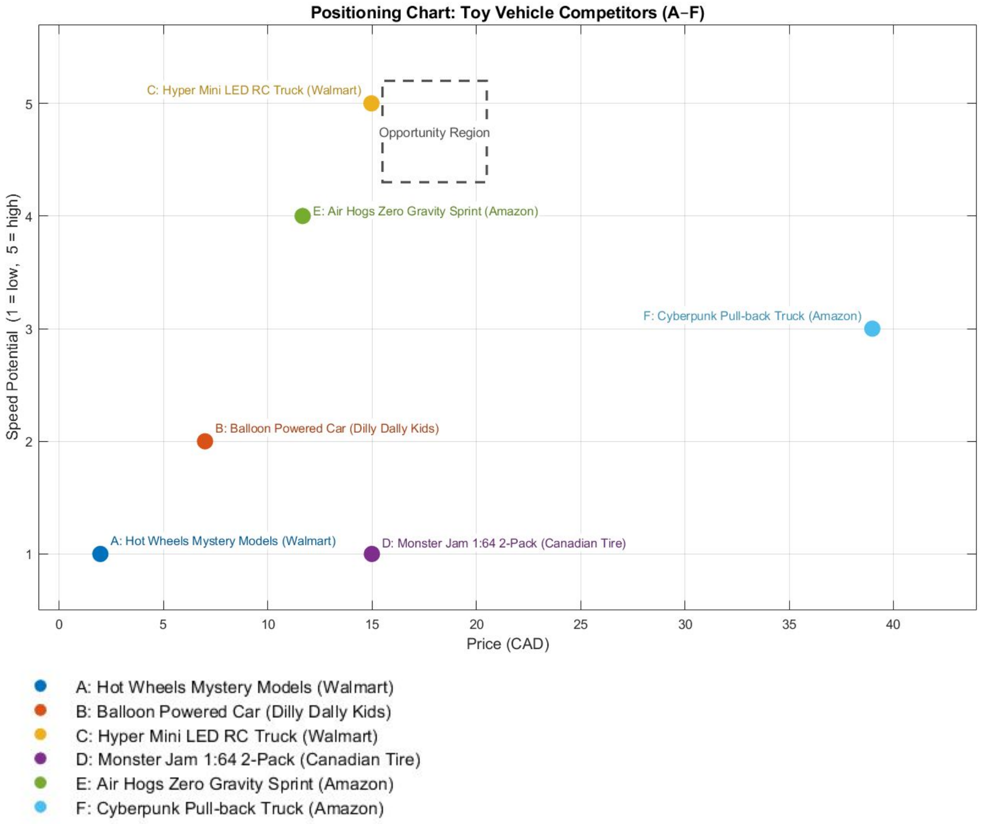
- Low-cost low performance
- ~$12–15 feature-driven (RC, novelty)
- High price styling, not speed

Key Insight

No product delivers **high speed at $15** without RC

The Opportunity

High speed + self-contained + $15 price point



Product Overview: *Wind-Runner*

A compact, propeller-powered sprint vehicle engineered to travel 20 ft in minimum time

Retail Price	Sprint Objective	Unit BOM Cost	Base Case NPV
$15 CAD	20 ft	$6.00	~$1.7M

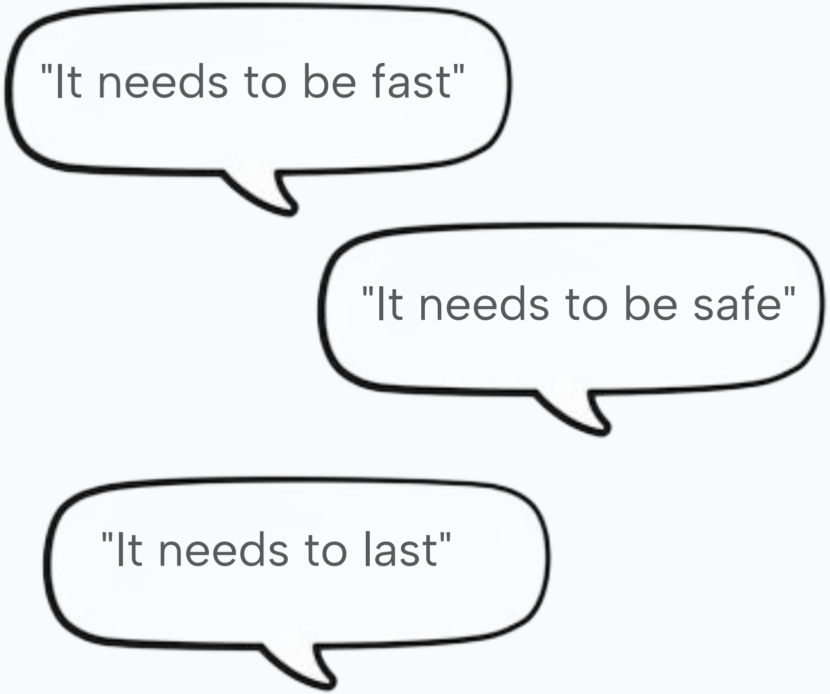
Key Requirements

- Minimize travel time over a fixed 20-foot (6.1m) sprint
- Must be self-contained
- Safe for Children (VOC driven)
- Durable (VOC Driven)
- Exciting (VOC Driven)

01

Design Methodology & Selection

VOC Driven Design



Safety
Mesh Guard (≤ 4mm gap), no exposed blades.

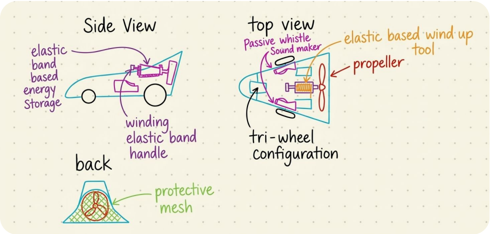
Durability
Foam bumper; withstands > 50 impacts at terminal speed.

Performance
Travel 20 ft in less than 15 seconds.

Cost
Retail < $15; Unit Cost < $6.

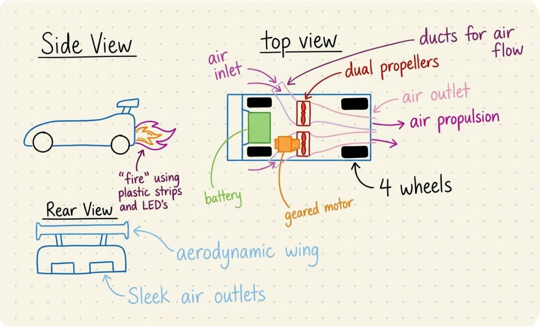
Excitement
LED + Sound; visually distinctive design.

Concept Generation



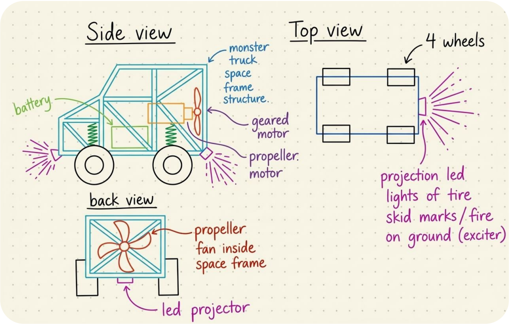
Rubber Band Racer

Mechanical energy storage using winding elastic bands for high torque.



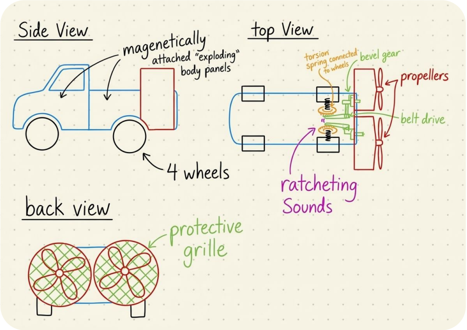
Ducted Fan Racer

Aerodynamic propulsion system with dual propellers and safety mesh.



Space Frame Monster Truck

High-clearance design featuring a gearless motor for rugged performance.



Ratchet Based Launcher

Spring-loaded mechanism with a ratchet release for high-speed starts.

Concept Selection

Concept 1: Rubber Band Racer

✔ Pro

Low cost, no electronics

✘ Cons

Speed variability and fatigue

Score: 3.55/5

Concept 2: Ducted Fan Racer

✔ Pro

High safety, airflow control

✘ Cons

Heavy, costly mold

Score: 3.8/5

Concept 3: Space Frame Monster Truck

✔ Pro

Consistent thrust, LED exciter

✘ Cons

Higher BOM

Score: 4.10/5

SELECTED

Concept 4: Ratchet Launcher

✔ Pro

Self contained high torque

✘ Cons

Complex, spring fatigue

Score: 3.45/5

Refined Concept: Optimized Fusion

Motor core (C3) + Partially ducted guard (C2) + Foam bumper (C1) + Whistle sound (C4)

Best balance of speed (4.10/5), safety, durability & cost for $15 mid-tier market

02

Experimental Optimization (DOE)

Full Factorial Design

Design Choice

Utilized a 2^3 Full Factorial Design, resulting in **8 unique treatment combinations**.

Replication & Power

Conducted **2 replicates per run** (16 total runs) to calculate pure error and ensure statistical significance.

Rationale

Chosen over "One-Factor-at-a-Time" (OFAT) to identify **hidden interactions** between factors that would otherwise be missed.

Noise Integration

Repeated across two surface levels (Tile/Carpet) to satisfy **Taguchi Robustness** criteria.

Run #	A: Blade Direction	B: Battery Type	C: Wheel Diameter
1	CW	NiMH	35 mm
2	CCW	NiMH	35 mm
3	CW	Alkaline	35 mm
4	CCW	Alkaline	35 mm
5	CW	NiMH	55 mm
6	CCW	NiMH	55 mm
7	CW	Alkaline	55 mm
8	CCW	Alkaline	55 mm

Significant Factors & Interactions

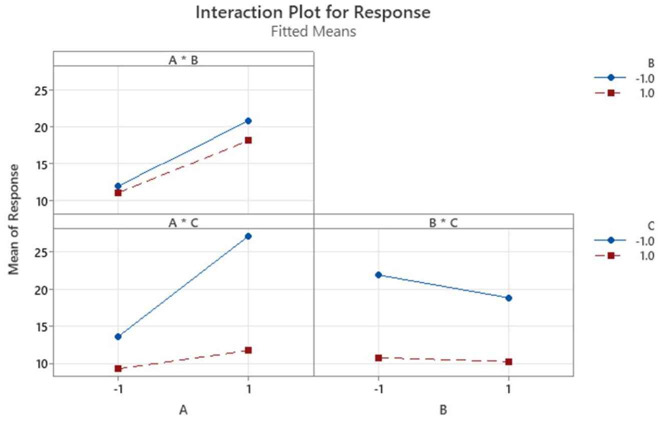
Significant Factors

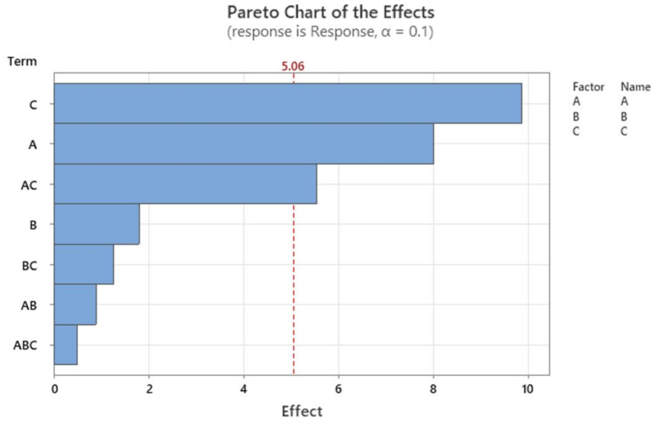
As seen in the pareto chart, the most significant control factors are **Wheel Diameter (C)** and **Propeller Direction (A)**.

The **interaction (AC)** between the two is also a statistically significant factor.

AC Interaction: Blade x Wheels

- Significant non-parallelism observed between Blade Direction (A) and Wheel Diameter (C).
- The effect of blade rotation is dependent on wheel geometry, likely due to spatial proximity.





Robust Design

Conducted experiments under different environmental conditions (carpet and tile) to simulate real-world variability. Evaluated both **performance (mean)** and **consistency (std. deviation)**.

Factor A

- Strong performance effect (level 1 → 2)
- High variability: **Less Robust**

Factor B

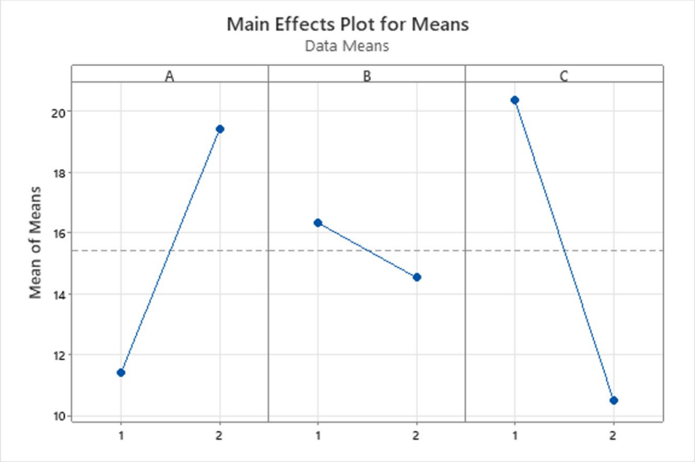
- Moderate mean decrease
- **More consistent** performance

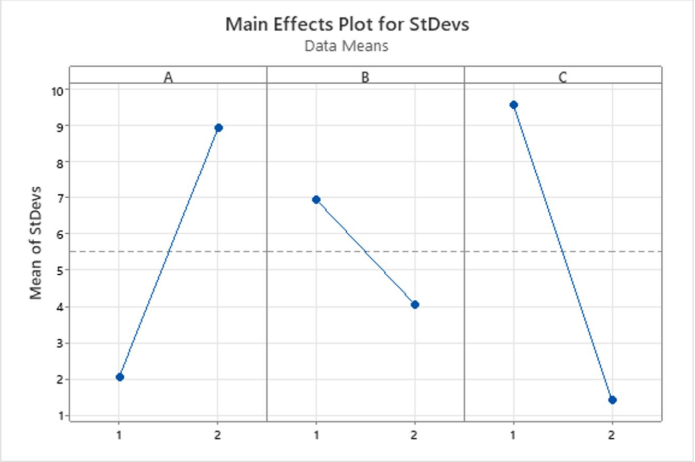
Factor C: Most Robust

Significant decrease in mean, but **large reduction in variability**. The primary driver for system stability.

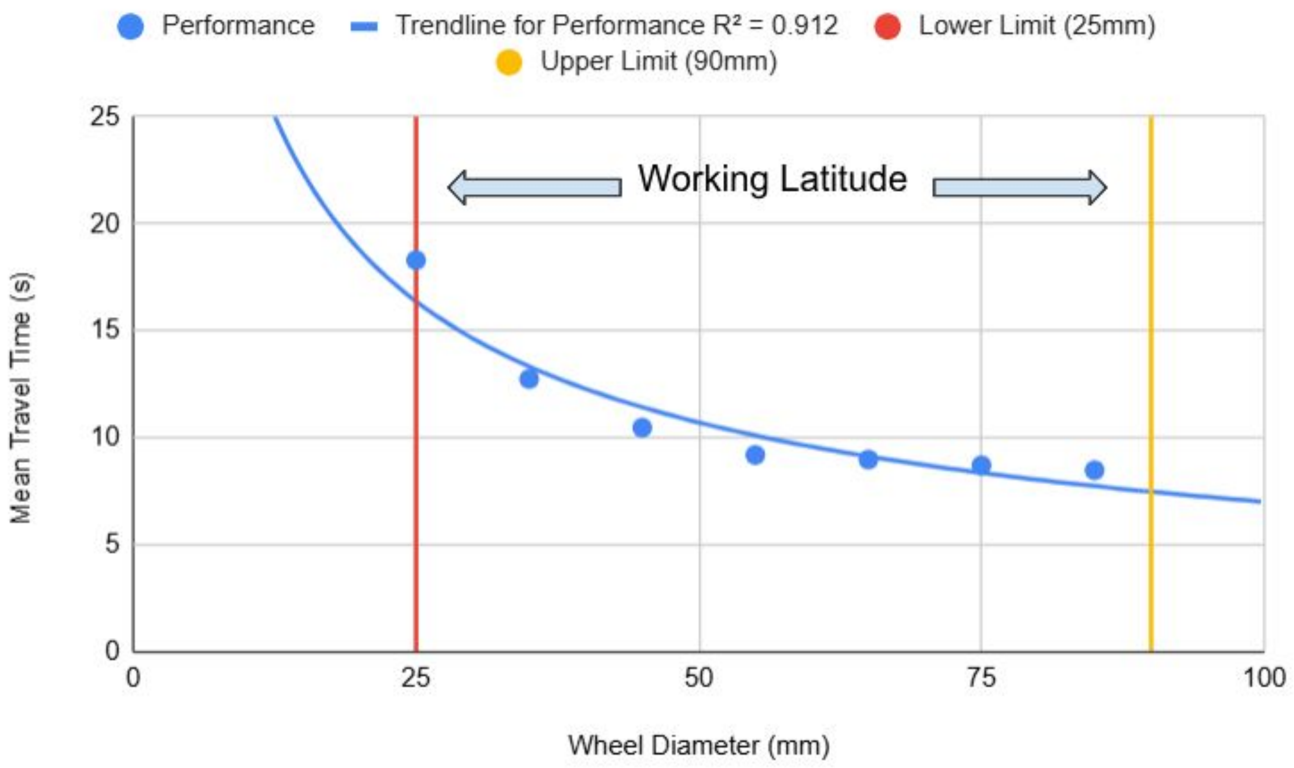
Design Optimization Principle

Prioritize factors that **reduce variability** while maintaining acceptable performance levels.





Latitude Development: Wheel Diameter



Tolerance Analysis

Component Tolerance Breakdown

Component (t_i) Nominal ± Tol

Shaft Bore Radius (t_1) 1.01 mm ± 0.15 mm

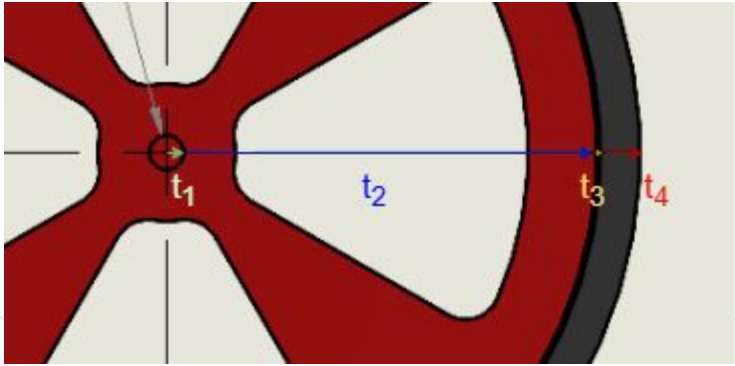
Wheel Body Radius (t_2)38.99 mm ± 0.25 mm

Wheel-to-Tire Gap (t_3) 0.20 mm ± 0.10 mm

TPU Tire Thickness (t_4) 2.30 mm ± 0.15 mm

Total Effective Radius: 42.50 mm ± 0.65 mm*

**Total tolerance shown is Worst-Case (WOW)*



Worst-Case Analysis (WOW)

$\Sigma|\sigma\_i| = 0.15 + 0.25 + 0.10 + 0.15 = 0.65$

- Radius tolerance: ±0.65 mm
- Diameter tolerance: ±1.30 mm
- **Latitude Used: 4.33% of 60 mm window**

Statistical Analysis (RSS)

$\sqrt{(\Sigma\ \sigma\_i^2)} = \sqrt{0.1175} = 0.343$

- Radius tolerance: ±0.343 mm
- Diameter tolerance: ±0.686 mm
- **Latitude Used: 2.29% of 60 mm window**

Design is **highly robust**: even worst-case tolerances consume only 4.33% of available functional latitude.

03

Final Design Validation

Optimal Configuration & Verification Run

A_1

CW Blade Direction

Reduces mean time by **8.01 s** vs CCW

S/N Delta: 6.18 dB

B_2

Alkaline Battery

Marginal **1.80 s** Improvement

S/N Delta: 3.23 dB

C_2

Big Wheel (85 mm)

Dominant Factor: Saves **9.87 s**

S/N Delta: 9.16 dB

Metric	Taguchi Additive Prediction	Factorial Reduced Prediction	Verification Measured
Mean Travel Time (s)	5.60	9.264	8.485
S/N Ratio (dB)	22.768 dB	–	22.214
Carpet-Tile Spread (s)	–	–	0.930

Verification confirms: S/N within **2.4%** of prediction; mean within **9.2%**.

04

Conclusion

Conclusion

Part A: Business Case

- DC Motor Performance Variant selected (4.10/5)
- Physics model validates ≤2 s target feasibility
- NPV = $1.71M @ 10% discount rate
- PFD, DFMEA and P–Diagram established

Part B: Optimization

- Optimal config: CW blade · Alkaline · Big wheel
- Big wheel is dominant (delta: 9.87 s, S/N 9.16 dB)
- S/N verified within 2.4%; mean within 9.2%
- 31× more surface–consistent than worst case

Latitude: Design Robustness

- Functional window: 25–85 mm (60 mm)
- Design point at 55 mm: ±30 mm safety margin
- No new risk from manufacturing variation

Tolerance: Manufacturing

- WOW = ±1.30 mm diameter (4.33% of latitude)
- RSS = ±0.69 mm (2.29% of latitude)
- Form, fit & function verified: robust design confirmed

Summary: **Robust Design Confirmed** with verified performance, high optimization, and minimal tolerance risk.

Next Steps

01. Demonstrate

Demonstrate rig performance

02. Validate

Validate sprint time vs. competitors

03. Refine

Refine DFMEA with test data

04. Confirm

Confirm supplier BOM quotes

05. Finalize

Finalize design freeze & CAD

Thank you!

Year & Quarter	Dev/Eng Cost	Tooling	Production Cost (COGS)	Marketing (12% Vol)	Revenue	Net Cash Flow (C_i)	Discounted Cash Flow ($\frac{C_i}{(1+r)^i}$)
Year 1: Q1	-$80,000	-$100,000	$0	$0	$0	**-$180,000**	-$175,610
Year 1: Q2	-$80,000	$0	$0	$0	$0	**-$80,000**	-$76,145
Year 1: Q3	-$80,000	$0	$0	$0	$0	**-$80,000**	-$74,288
Year 1: Q4	-$80,000	$0	-$190,000	$0	$0	**-$270,000**	-$244,606
Year 2: Q5	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$204,330
Year 2: Q6	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$199,346
Year 2: Q7	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$194,484
Year 2: Q8	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$189,741
Year 3: Q9	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$185,113
Year 3: Q10	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$180,598
Year 3: Q11	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$176,193
Year 3: Q12	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$171,896
Year 4: Q13	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$167,703
Year 4: Q14	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$163,613
Year 4: Q15	-$50,000	$0	-$190,000	-$3,800	$475,000	**+$231,200**	+$159,622
Year 4: Q16	-$50,000	$0	$0	-$3,800	$475,000	**+$421,200**	+$283,731
PROJECT TOTALS	**-$920,000**	**-$100,000**	**-$2,280,000**	**-$45,600**	**+$5,700,000**	**+$2,354,400**	**+$1,705,724**